

Infrastructure Governance: Compliance & Threshold Manual

This document provides the formal mathematical and engineering foundations for the ShineGuard IoT platform. All thresholds have been calibrated against global industrial standards to ensure urban resilience and hardware longevity.

1. Electrical & Telemetry Thresholds

Metric	Nominal Range	Warning Level	Critical Trip	Compliance Standard
Voltage	210V - 240V	< 200V / > 250V	< 180V	IEC 60038
Temperature	20°C - 55°C	> 65°C	> 85°C	Industrial Grade
Current (A)	0.01A - 2.0A	> 2.5A	> 5.0A	IEC 60598-1
Ambient Light	10 - 1000 lux	< 10 lux	0 lux	EN 13201
Humidity	10% - 60%	> 80%	> 95%	IP65/67 Standard

2. Cybersecurity & Governance

Rule Name	Operational Limit	Standard Alignment	Security Justification
API Velocity	60 req / min	OWASP API7:2023	DDoS Mitigation
Auth Expiry	15 mins Idle	NIST SP 800-63B	Hijacking Defense
Bulk Lockout	5 Min Window	High-Assurance IAM	Multi-factor Re-Auth
Log Chain	HMAC-SHA256	FIPS 180-4	Non-Repudiation

3. Asset Lifecycle Thresholds

Lifecycle Event	Reference Value	Maintenance Category
Service Interval	12 Months / 4k Hrs	Preventative (EN 13306)
Health Threshold	> 5 Critical Errors	Condition-Based Maintenance
Resolution Target	< 24 Hours	SLA Compliance (ISO 9001)
Retirement Age	5 Years / 25k Hrs	Total Cost of Ownership (TCO)